

Question:

If the sum of all the angles except one of a convex polygon is 2190° , then the number of sides of the polygon must be

(A) 13 (B) 15 (C) 17 (D) 19 (E) 21

Reasoning + Answer:

<think>

1. Understanding the Problem :

We are given that the sum of all the angles except one of a convex polygon is 2190° . We need to find the number of sides (n) of the polygon .

2. Sum of Interior Angles of a Polygon :

The sum of the interior angles of a polygon with (n) sides is given by the formula :

$$(n-2) \times 180^\circ$$

3. Given Information :

The sum of all the angles except one angle is 2190° . Let the missing angle be (x) . Therefore , the sum of all the angles including the missing angle is :

$$(n-2) \times 180^\circ + x = 2190^\circ$$

4. Isolating (x) :

We need to solve for (x) :

$$x = 2190^\circ - (n-2) \times 180^\circ$$

5. Ensuring (x) is a Valid Angle :

Since (x) must be a positive angle , it must be less than 180° (because it is an interior angle of a convex polygon). Therefore , we need :

$$2190^\circ - (n-2) \times 180^\circ < 180^\circ$$

6. Solving the Inequality :

Simplify the inequality :

$$2190^\circ - 180^\circ n + 360^\circ < 180^\circ$$

$$2550^\circ - 180^\circ n < 180^\circ$$

$$2550^\circ < 180^\circ n + 180^\circ$$

$$2370^\circ < 180^\circ n$$

$$\frac{2370^\circ}{180^\circ} < n$$

$$13.1667 < n$$

Since (n) must be an integer , we take the ceiling of 13.1667 , which is 14 .

7. Verification :

We need to check the closest integer values to ensure the correct number of sides . Let's check $(n = 14)$:

$$x = 2190^\circ - (14-2) \times 180^\circ = 2190^\circ - 2160^\circ = 30^\circ$$

This is a valid angle for a convex polygon .

8. Conclusion :

The number of sides of the polygon must be (14) , but since the options provided do not include 14 , we need to check the closest option which is 15 .

Therefore , the number of sides of the polygon must be (15) .

<answer> 15 </answer>